



# Consent Content Management System

|                              |                                                                             |
|------------------------------|-----------------------------------------------------------------------------|
| <b>Project/Feature Name:</b> | Consent Content Management System                                           |
| <b>Product Owner:</b>        | Casey Duld                                                                  |
| <b>Project Manager:</b>      | Adam Antolik                                                                |
| <b>Figma Designs:</b>        | [Link]                                                                      |
| <b>Jira ticket:</b>          | <a href="#">[PMO-45] Consent Refinement &amp; Systemic Integrity - Jira</a> |
| <b>Document Version:</b>     | 1.0                                                                         |
| <b>Status:</b>               | Draft                                                                       |

## 1. Executive Summary

The purpose of this project is build a platform that stores and maintains all consent language in a centralized location to be utilized both internally and externally in our ordering systems and serves as a source of truth for consent language. The platform will allow for downstream integration so that when changes are needed, all ordering systems are updated to ensure consistency across systems.

## 2. Background & Context

### 2.1 Problem Statement

With modifications performed over time to consent language, inconsistencies in language, response logic, and data mapping have contributed to operational inefficiencies, reporting challenges, and reduced confidence in consent interpretation and use.

### 2.2 Current State

Variations in how consents are titled, displayed, stored, and mapped, combined with limited structured storage of historical consents, have led to manual corrections, inefficiencies, and compliance risk across accessioning, reporting, medical affairs, and research workflows. Without a standardized, end-to-end consent framework, Baylor Genetics lacks a reliable and

scalable foundation to ensure consent data is clearly understood, accurately applied, and consistently used across the testing lifecycle, creating the need for coordinated refinement of consent language, logic, and system integration

### 2.3 Target Audience

Primary Stakeholders:

1. Product team
2. Compliance
3. Reporting team
4. Accessioning (SMD) team
5. Medical Affairs
6. Research & Development team

Secondary Stakeholders:

1. Client Services team
2. Genetic Counselor support team
3. Commercial team

### 3. Goals & Success Metrics

#### Business Goals

- Establish a final, authoritative platform for consent language (audit all information before adding to the system for a true up).
- Expose APIs for systems to read master data from the repository platform.
- Update portal, paper and EPIC consents based on finalized language.

#### Key Results & Metrics (OKRs)

| Objective                                                                              | Key Result                                                      |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Remove data discrepancies between consent language across ordering systems             | 100% language alignment between Portal, Paper and Epic          |
| All consent language is mapped to respective test codes and responses captured in each | 100% of consent language is stored in content management system |

|                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| ordering system are fully documented                                                |                                                                                      |
| All BG systems/teams are using the content management system for consent attributes | 100% of systems/teams are using the content management system as the source of truth |

## 4. Functional Requirements

### 4.1. Data Structure

These requirements define how consent data is structured, stored, and validated within the content management system.

| ID   | Requirement                                                                                                                                                                                                                                                                                                                                 | Priority |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| DS-1 | The system must ensure that every consent language has all required attributes stored in a structured format, including the following data fields: Title, Consent Language, System, Test Code, Test Name, Test Category, Expected Response, Acceptable Responses, Response Default, Effective Date, Retirement Date, Active/Retired Status. | Must     |
| DS-2 | The system must support multiple language versions per consent category to accommodate different ordering systems, including historical tracking of consent updates.                                                                                                                                                                        | Must     |
| DS-3 | The system must support effective dating for consent language.                                                                                                                                                                                                                                                                              | Must     |
| DS-4 | The system must support relationships between consents and test codes and all associated attributes required for downstream system functionality (refer to Data Fields section below).                                                                                                                                                      | Must     |

|      |                                                                                                                                                                                                                                                                                                          |      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| DS-5 | The system must enforce mandatory fields before a test code can be saved or activated. Required fields include: Title, Consent Language, System, Test Code, Test Name, Test Category, Expected Response, Acceptable Responses, Response Default, Effective Date, Retirement Date, Active/Retired Status. | Must |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|

## 4.2. Governance

These requirements ensure data integrity, control, auditability, and proper oversight of consent changes.

| ID    | Requirement                                                                                                                              | Priority |
|-------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| GOV-1 | The system must maintain full auditability of all consent language, including who made changes, what changed, and when changes occurred. | Must     |
| GOV-2 | The system must maintain version control for all consent language, including historical snapshots and previous values.                   | Must     |
| GOV-3 | The system must require approval before any existing consent language changes can be activated.                                          | Must     |
| GOV-4 | The system must support role-based access control, restricting permissions based on user roles (refer to Role Types section below).      | Must     |
| GOV-5 | The system must prevent unapproved or incomplete consent language from being made active or available downstream.                        | Must     |
| GOV-6 | The system must require all mandatory fields to be completed prior to adding a new test code to the system. Mandatory fields             | Must     |

|       |                                                                                                                                                |      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|------|
|       | include: Title, Consent Language, System, Test Code, Test Category, Expected Response, Acceptable Responses, Response Default, Effective Date. |      |
| GOV-7 | The system must maintain a complete audit trail of approvals, including approver identity and timestamps.                                      | Must |
| GOV-8 | The system will utilize SSO (single sign on) for all registered users.                                                                         | Must |

### 4.3. Workflow

These requirements define how users initiate, route, approve, and implement consent changes, including governance enforcement and approval routing.

| ID   | Requirement                                                                                                                                                                                                                   | Priority |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| WF-1 | The system must support a structured approval workflow: Draft → Intake Submitted → Review → Approved Pending Implementation → Active.                                                                                         | Must     |
| WF-2 | The system must require submission through a formal intake process including: structured submission form, change type classification, required supporting documentation, and automatic assignment of a unique intake ID.      | Must     |
| WF-3 | The system must require all mandatory fields relevant to the requested change to be completed prior to submission for approval. Mandatory fields include: Title, Consent Language, System, Test Code, Test Category, Expected | Must     |

|      |                                                                                                                                                                                                  |        |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|      | Response, Acceptable Responses, Response Default, Effective Date.                                                                                                                                |        |
| WF-4 | The system must be able to push consent language updates to digital ordering platforms once approved (refer to Approval Routing section below).                                                  | Should |
| WF-5 | The system automatically sends an email notification to marketing team when an update is needed for paper TRFs and stand-alone consents once approved (refer to Approval Routing section below). | Should |

#### 4.4 Approval Routing

| ID   | Requirement                                                                                                                       | Priority |
|------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
| WF-6 | The system must generate and send automated approval notifications via email (Teams as a secondary) to the Consent Product Owner. | Must     |
| WF-7 | The approver should be able to approve the request without having to log back into the system.                                    | Should   |
| WF-8 | The system must capture structured electronic approvals, including: individual vote, timestamp and approval status.               | Must     |
| WF-9 | The system must automatically reject a request if the required approver records a veto and notify the Data Administrator.         | Must     |

|       |                                                    |      |
|-------|----------------------------------------------------|------|
| WF-10 | Until Approved, the changes will not be effective. | Must |
|-------|----------------------------------------------------|------|

#### 4.5 Versioning, Implementation & Validation

| ID    | Requirement                                                                                                          | Priority |
|-------|----------------------------------------------------------------------------------------------------------------------|----------|
| WF-11 | The system must automatically generate a new Consent version upon approval of changes.                               | Must     |
| WF-12 | The system must archive prior versions and maintain a complete historical record, including field-level change logs. | Must     |
| WF-13 | The system must apply effective date logic to all approved changes.                                                  | Must     |

#### 4.6 Communication & Distribution

| ID    | Requirement                                                                                                                                                                                                                                                                                                       | Priority |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| WF-14 | The system must generate automated email change summaries upon approval and implementation of updates. The content must include the consent title and associated attribute that was updated (including the previous language and new language) and what date the change was effective. Format can be in an email. | Must     |
| WF-15 | The system must notify all registered users, regardless of                                                                                                                                                                                                                                                        | Must     |

|       |                                                                                                           |      |
|-------|-----------------------------------------------------------------------------------------------------------|------|
|       | access level, of any changes made in the system.                                                          |      |
| WF-16 | The system must allow export of updated consent language in Excel or PDF (e.g., Master Consent Language). | Must |

## 5. User Types

| User Type          | Key Responsibility                                                                                                                              | Permissions                                                                                                                                                                                                                                                                                                                          | Department                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Administrator      | Ensure the system infrastructure and security controls function properly. Manage system configuration and user access.                          | <ul style="list-style-type: none"> <li>• Full system access</li> <li>• Create, edit, archive, or delete consent records</li> <li>• Manage users, roles, and permissions</li> <li>• Configure workflows and integrations</li> <li>• Override workflows when required</li> <li>• Access full audit logs and version history</li> </ul> | Business Platforms<br>Technology                       |
| Data Administrator | Serve as the operational steward of the consent language catalog. Maintain the integrity and accuracy of the consent content management system. | <ul style="list-style-type: none"> <li>• Create and edit consent records</li> <li>• Maintain consent language attributes (e.g. Title, Consent Language, System, Test Code, Test Category, Effective Date)</li> <li>• Initiate and manage change requests</li> <li>• Publish approved updates</li> </ul>                              | Business Platforms<br>Technology<br>Product Management |



|        |                                                                                           |                                                                                                                                                                                                                                                                                                         |                          |
|--------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|        |                                                                                           | <ul style="list-style-type: none"> <li>• Retire or archive consent language</li> <li>• Manage consent language attribute structure</li> </ul>                                                                                                                                                           |                          |
| Viewer | Use the content management system as a reference source for accurate consent information. | <ul style="list-style-type: none"> <li>• View all consent records</li> <li>• Search and filter consent data</li> <li>• Export reports or reference materials</li> </ul> <p><b>Restrictions</b></p> <ul style="list-style-type: none"> <li>• No editing, approval, or publishing capabilities</li> </ul> | All internal departments |

## 6. Data Field Definitions [Consent Attributes]

Definitions for each consent attribute and categories associated with it to be reviewed and approved before adding it to the content management system to reduce discrepancies and have a data dictionary that is consistent.

| Column Name / Required Field | Definition/Purpose                                                                                                                                                  | Requirement                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Title                        | Stores the approved title for each consent section.                                                                                                                 | Must not exceed 50 characters   |
| Consent Language             | Stores the approved language for each consent section.                                                                                                              | Must not exceed 2500 characters |
| Test Code                    | Stores the test code designated to each test.                                                                                                                       | Must not exceed 15 characters   |
| Test Name                    | Stores the official name of a test to be utilized in all internal and external facing materials and communications (including required trademarks when applicable). | Must not exceed 100 characters  |

|                      |                                                                                                       |                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Category        | Stores the category a test will fall under for reporting purposes.                                    | <p>Must select from a pre-determined list of categories:</p> <ul style="list-style-type: none"> <li>o Bio</li> <li>o Carrier</li> <li>o CMA</li> <li>o Cyto</li> <li>o Mito</li> <li>o NGS</li> <li>o NIPT</li> <li>o Other</li> <li>o PGx</li> <li>o WES</li> <li>o WGS</li> </ul> |
| System               | Stores the language by system as there are subtle differences in language between ordering platforms. | <p>Must select from a pre-determined list of categories:</p> <ul style="list-style-type: none"> <li>o Portal</li> <li>o Paper</li> <li>o Epic</li> </ul>                                                                                                                            |
| Expected Response    | Stores whether the consent section is informational only versus requires a response upon order.       | <p>Must select from a pre-determined list of categories:</p> <ul style="list-style-type: none"> <li>o Yes</li> <li>o No</li> </ul>                                                                                                                                                  |
| Acceptable Responses | Stores the relevant options for responses based on test code (opt-in, opt-out, N/A).                  | <p>Must select from a pre-determined list of categories:</p>                                                                                                                                                                                                                        |

|                       |                                                                                                                                |                                                                                                                                                                           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                                                                                                | <ul style="list-style-type: none"> <li>o Opt-in</li> <li>o Opt-out</li> </ul>                                                                                             |
| Default Response      | Stores the default response when consent is unanswered (i.e. left blank).                                                      | <p>Must select from a pre-determined list of categories:</p> <ul style="list-style-type: none"> <li>o Opt-in</li> <li>o Opt-out</li> <li>o N/A</li> <li>o Null</li> </ul> |
| Effective Date        | Stores the date the version of consent went into effect is available to public on ordering systems.                            | Must be entered in the format MM/DD/YYYY                                                                                                                                  |
| Retirement Date       | Stores the date when the consent language is retired and no longer applicable.                                                 | Must be entered in the format MM/DD/YYYY                                                                                                                                  |
| Active/Retired Status | Stores if the consent language is active or retired. Active language is defined as included in orderable and reportable tests. | Must choose Active or Retired                                                                                                                                             |

## 7. Non-Functional Requirements

| Category    | Requirement                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalability | Should be able to support simultaneous and bulk updates (50+)                                                                                            |
| Scalability | The system should allow Data Administrators to add, edit, or archive data fields (columns) as business needs evolve (refer to User Types section below). |
| Flexibility | The system should support a flexible data model that allows                                                                                              |



Primary Contents Categories:

Test Information  
(Variation A)

- Version
- 2.1
  - 2
  - 1
- Status
- Live
  - Approved
  - Pending
  - Archived

Test Information  
(Variation B)

Test Information  
(Variation C)

Conceptually  
Organized By:

Testing Categories  
(e.g., Whole Genome Sequencing)

Applied To:

Specific Test Code(s)  
(e.g. 1500)

Presentation  
Context:

Ordering Portal  
(Digital)

Specific Test Requisition Form  
(Paper)

Stand Alone Consent Form  
(Paper)

EHR  
(Digital)

Test Reporting Result  
(Variation A)

Test Reporting Result  
(Variation B)

Test Reporting Result  
(Variation C)

Test Reporting Result  
(Variation D)

Additional Reporting  
(Variation A)

Additional Reporting  
(Variation B)

Considerations and Limitations  
(Variation A)

Considerations and Limitations  
(Variation B)

Considerations and Limitations  
(Variation C)

Confidentiality and Sample Rate  
(Variation A)

Consent Key with Responses  
(Variation A)

Financial Agreement  
(Variation A)

Consent Key with Responses  
(Variation B)

Financial Agreement  
(Variation B)

Additional Consents:

For Samples from New York State Residence  
(Variation A)

Additional Reporting: Incidental Findings  
(Variation A)

Additional Reporting: ACMG Secondary Findings  
(Optional for comparators)  
(Variation A)

Reporting of Category I (ACMG) Secondary Findings  
(Variation A)

Use of Data and Specimen for Research Purposes  
(Variation A)



|                    |                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | addition of new attributes without requiring schema changes.                                                                                                      |
| Downstream systems | If there is a downstream workflow that is dependent on the consent field update then that downstream workflow should continue to work as it is currently designed |

## 8. Risks

### Risks & Mitigation

| Risk                                                                          | Probability | Impact | Mitigation Strategy                                                                                                                       |
|-------------------------------------------------------------------------------|-------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Single source of truth is unclear or not honored                              | Low         | High   | Formal designation of Content Management System as the <i>system of record</i> for all consent language and definitions written in a SOP. |
| Content Management System does not align with future digital consent platform | Low         | High   | Design Content Management System content objects to be channel-agnostic (paper, portal, digital assistant)                                |

## 9. Out of Scope

- Interface with other modules (test code single source of truth) within content management system

- System requirements to automatically generate a consent document (i.e. consent builder)
- Mechanism to retract outdated paper consents being used in the field.

## 10. Revision History

| Version | Date       | Author       | Changes       |
|---------|------------|--------------|---------------|
| 1.0     | 04/28/2026 | Leslie Cohen | Initial draft |